

11.3 EVD of Non-symmetric Square Matrices



Core skills you will master in this section:

- ▶ For the EVD of a non-symmetric matrix, V is no longer an orthogonal matrix.
- ▶ V and its inverse matrix (if it exists) cancel each other's transformation effect; the product of their determinants is 1.
- ▶ This section decomposes V into shear and scaling operations.
- ▶ The EVD decomposition result of the non-symmetric matrix in this section can be viewed as "scale $\rightarrow$ shear $\rightarrow$ scale $\rightarrow$ shear $\rightarrow$ scale."
- ▶ A uniform scaling matrix: every direction is an eigen-direction.
- ▶ The geometric meaning of non-uniform scaling: each direction has a different stretch factor.
- ▶ Rotation matrices: when complex numbers appear, real-number diagonalization is usually not possible.
- ▶ An eigenvalue of 0 means the linear transformation involves a dimensionality reduction.
- ▶ A negative eigenvalue means a reflection.
- ▶ Even with eigenvalues 0 and 1 present, a square matrix may still be diagonalizable.

In the previous section, we saw that the eigenvalue decomposition of a real symmetric matrix corresponds to "rotate $\rightarrow$ scale $\rightarrow$ rotate"; however, the geometric operations corresponding to the eigenvalue decomposition of many non-symmetric matrices are not so obvious. This section uses an example to show the geometric operations of the eigenvalue decomposition (EVD) of a non-symmetric matrix. Of course, this is just one of many cases, and more scenarios will be shown at the end of this section.

The EVD of a Non-symmetric Matrix

Given the following non-symmetric square matrix A

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$$

The eigenvalue decomposition of the square matrix A is

$$A = \underbrace{V}_{\begin{bmatrix} 1 & -\sqrt{2}/2 \\ 0 & \sqrt{2}/2 \end{bmatrix}} \underbrace{A}_{\begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}} \underbrace{V^{-1}}_{\begin{bmatrix} 1 & 1 \\ 0 & \sqrt{2} \end{bmatrix}} = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$$

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Code and PDF file downloads: <https://github.com/Visualize-ML>

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Comments and corrections are welcome. The book's dedicated email address: jiang.visualize.ml@gmail.com

In the equation above, $\mathbf{V}$'s column vectors are still unit vectors, but they are not orthogonal.

⚠ Note that in the equation above, the rightmost matrix is the inverse of $\mathbf{V}$. Since $\mathbf{V}$ is not an orthogonal matrix, the inverse of $\mathbf{V}$ cannot be written as $\mathbf{V}^T$.

Compute $\mathbf{V}$'s Gram matrix $\mathbf{V}^T @ \mathbf{V}$

$$\mathbf{V}^T @ \mathbf{V} = \begin{bmatrix} 1 & 0 \\ -\sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix} \begin{bmatrix} 1 & -\sqrt{2}/2 \\ 0 & \sqrt{2}/2 \end{bmatrix} = \begin{bmatrix} 1 & -\sqrt{2}/2 \\ -\sqrt{2}/2 & 1 \end{bmatrix}$$

Writing $\mathbf{V}$ as $[\mathbf{v}_1, \mathbf{v}_2]$, the matrix multiplication above can be expanded as

$$\mathbf{V}^T @ \mathbf{V} = \begin{bmatrix} \mathbf{v}_1^T \\ \mathbf{v}_2^T \end{bmatrix} \begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{v}_1^T \mathbf{v}_1 & \mathbf{v}_1^T \mathbf{v}_2 \\ \mathbf{v}_2^T \mathbf{v}_1 & \mathbf{v}_2^T \mathbf{v}_2 \end{bmatrix}$$

Comparing this equation with the previous one, we find that the diagonal elements are 1, which means $\mathbf{v}_1$ and $\mathbf{v}_2$ are unit vectors; clearly $\mathbf{v}_1$ and $\mathbf{v}_2$ are not orthogonal — in this example, the angle between them is 135 degrees.

And $\mathbf{V}$'s inverse matrix is

$$\mathbf{V}^{-1} = \begin{bmatrix} 1 & 1 \\ 0 & \sqrt{2} \end{bmatrix}$$

We find that $\mathbf{V}^{-1}$'s column vectors are not all unit vectors.

Diagonalizable

Fortunately ("luckily"), in the equation above, $\mathbf{V}$'s column vectors are linearly independent, so the square matrix $\mathbf{V}$ has full rank; in other words, $\mathbf{V}$ is invertible.

This means $\mathbf{A}$ can be diagonalized, that is

$$\mathbf{A} = \underbrace{\mathbf{V}^{-1}}_{\mathbf{V}} \underbrace{\mathbf{A}}_{\mathbf{A}} \underbrace{\mathbf{V}}_{\mathbf{V}^{-1}} = \begin{bmatrix} 1 & 1 \\ 0 & \sqrt{2} \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -\sqrt{2}/2 \\ 0 & \sqrt{2}/2 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$$

Once a matrix has been successfully diagonalized, many matrix-related computational problems are significantly simplified.

For example, computing the determinant no longer requires a tedious expansion — it can instead be obtained directly as the product of the elements on the main diagonal of the diagonal matrix. For instance, the determinant of matrix $\mathbf{A}$ equals the product of its eigenvalues:

$$\det(\mathbf{A}) = \prod_{j=1}^D \lambda_j$$

Besides the determinant, diagonalization also greatly simplifies computing powers of a matrix.

If $\mathbf{A}$ can be diagonalized, the square of matrix $\mathbf{A}$ can be written as:

$$A^2 = VAV^{-1}VAV^{-1} = VA^2V^{-1} = V \begin{bmatrix} (\lambda_1)^2 & & & \\ & (\lambda_2)^2 & & \\ & & \ddots & \\ & & & (\lambda_D)^2 \end{bmatrix} V^{-1}$$

Similarly, A 's n -th power can be written as:

$$A^n = VAV^{-1} \dots VAV^{-1} = VA^nV^{-1} = V \begin{bmatrix} (\lambda_1)^n & & & \\ & (\lambda_2)^n & & \\ & & \ddots & \\ & & & (\lambda_D)^n \end{bmatrix} V^{-1}$$

Next, let's again use a geometric perspective to help us understand A , that is, A 's eigenvalue decomposition and the corresponding geometric operations.

The Non-symmetric Matrix Acting on e_1 and e_2

Let's first look at the geometric operation of matrix A on the unit vectors.

As shown in Figure 1, the action of the non-symmetric matrix A on e_1 corresponds to the matrix multiplication Ae_1

$$Ae_1 = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

The action of the non-symmetric matrix A on e_2 corresponds to the matrix multiplication Ae_2

$$Ae_2 = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

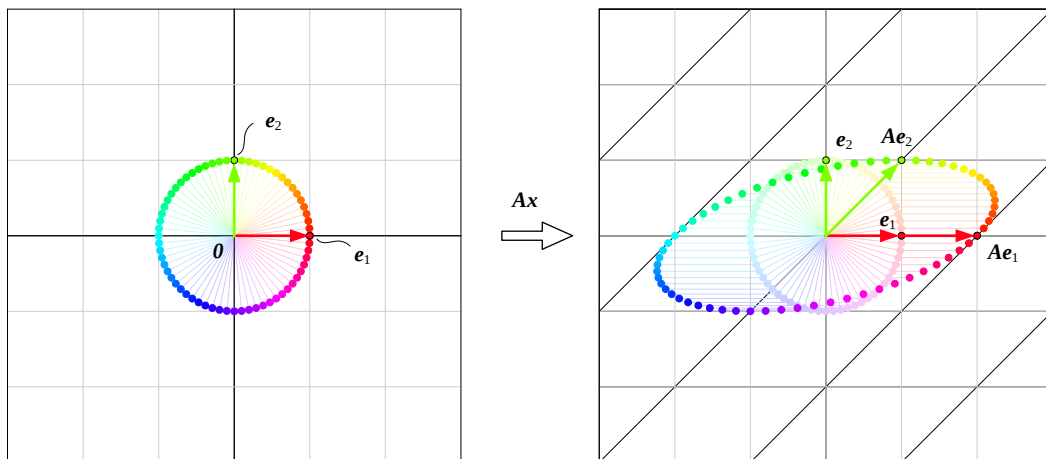


Figure 1. Linear Transformation of Matrix A on e_1 and e_2

Figure 2 shows the step-by-step linear transformation of the matrix product VAV^{-1} on e_1 and e_2 .

The action of the diagonal matrix A is still scaling — this is exactly the same as the action of the diagonal matrix in spectral decomposition.

We see that the action of V^{-1} and V on the shape appears to be "shear + scale"; let's analyze this in detail below.

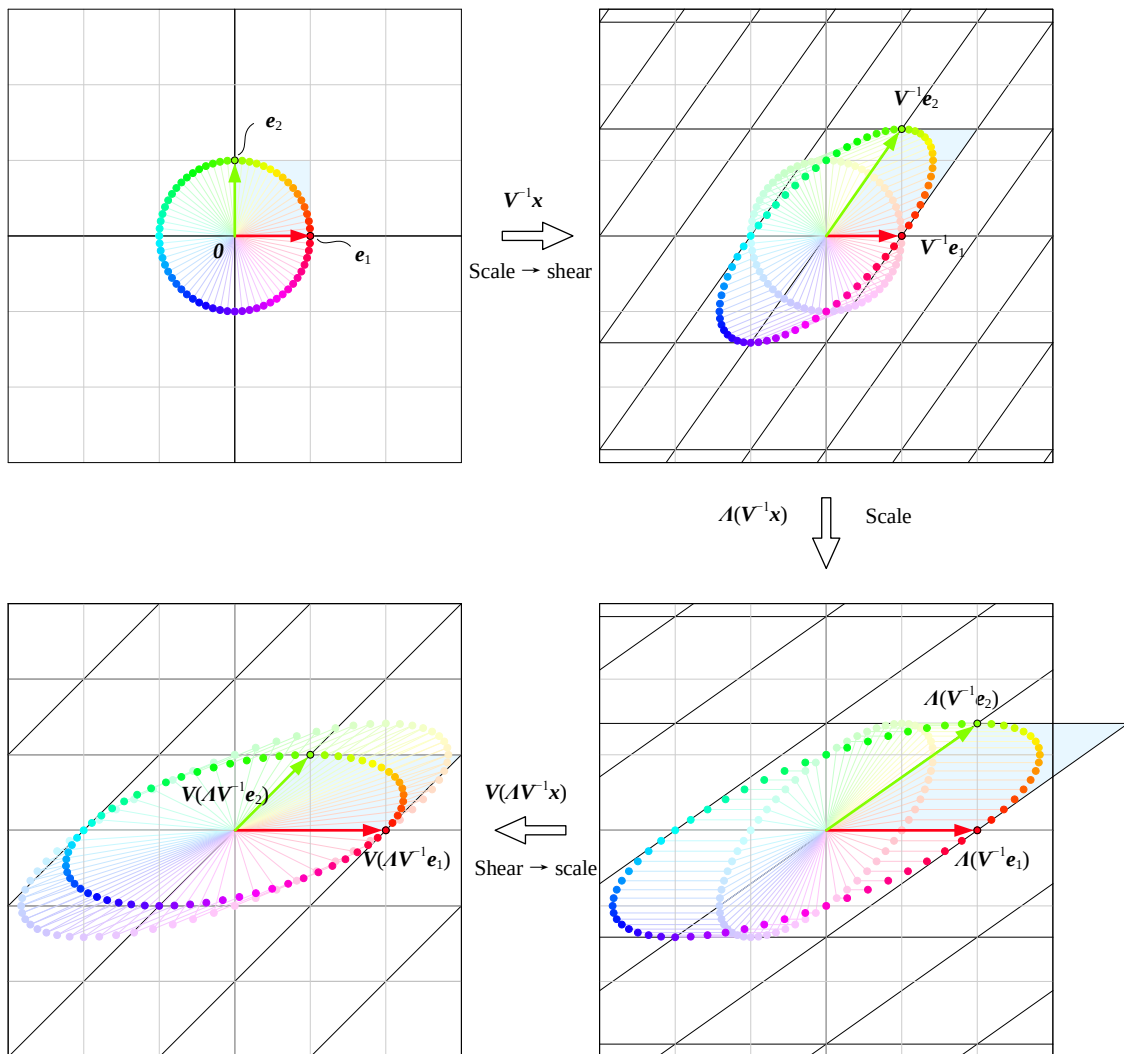


Figure 2. Linear Transformation of the Matrix Product VAV^{-1} on e_1 and e_2

Decomposing V

Looking closely at V , V can be decomposed into the product of two matrices

$$V = \begin{bmatrix} 1 & -\sqrt{2}/2 \\ 0 & \sqrt{2}/2 \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & \sqrt{2}/2 \end{bmatrix}}_{\text{Scale, } S} @ \underbrace{\begin{bmatrix} 1 & -\sqrt{2}/2 \\ 0 & 1 \end{bmatrix}}_{\text{Shear, } K} = S @ K$$

K is a shear along the horizontal axis

$$K = \begin{bmatrix} 1 & -\sqrt{2}/2 \\ 0 & 1 \end{bmatrix}$$

S is the scaling matrix

$$S = \begin{bmatrix} 1 & 0 \\ 0 & \sqrt{2}/2 \end{bmatrix}$$

To be precise, the equation above is a scaling along the vertical axis.

In the equation above, $Vx = SKx$ – the order of geometric operations on the column vector x is: shear first (K), then scale (S).

V^{-1} Acting on e_1 and e_2

This way, we can write V^{-1} as

$$V^{-1} = \begin{bmatrix} 1 & 1 \\ 0 & \sqrt{2} \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & \sqrt{2}/2 \\ 0 & 1 \end{bmatrix}}_{\text{Shear, } K^{-1}} @ \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & \sqrt{2} \end{bmatrix}}_{\text{Scale, } S^{-1}} = K^{-1}S^{-1}$$

As shown in Figure 3, V^{-1} 's action is equivalent to first scaling with S^{-1} , then shearing with K^{-1} .

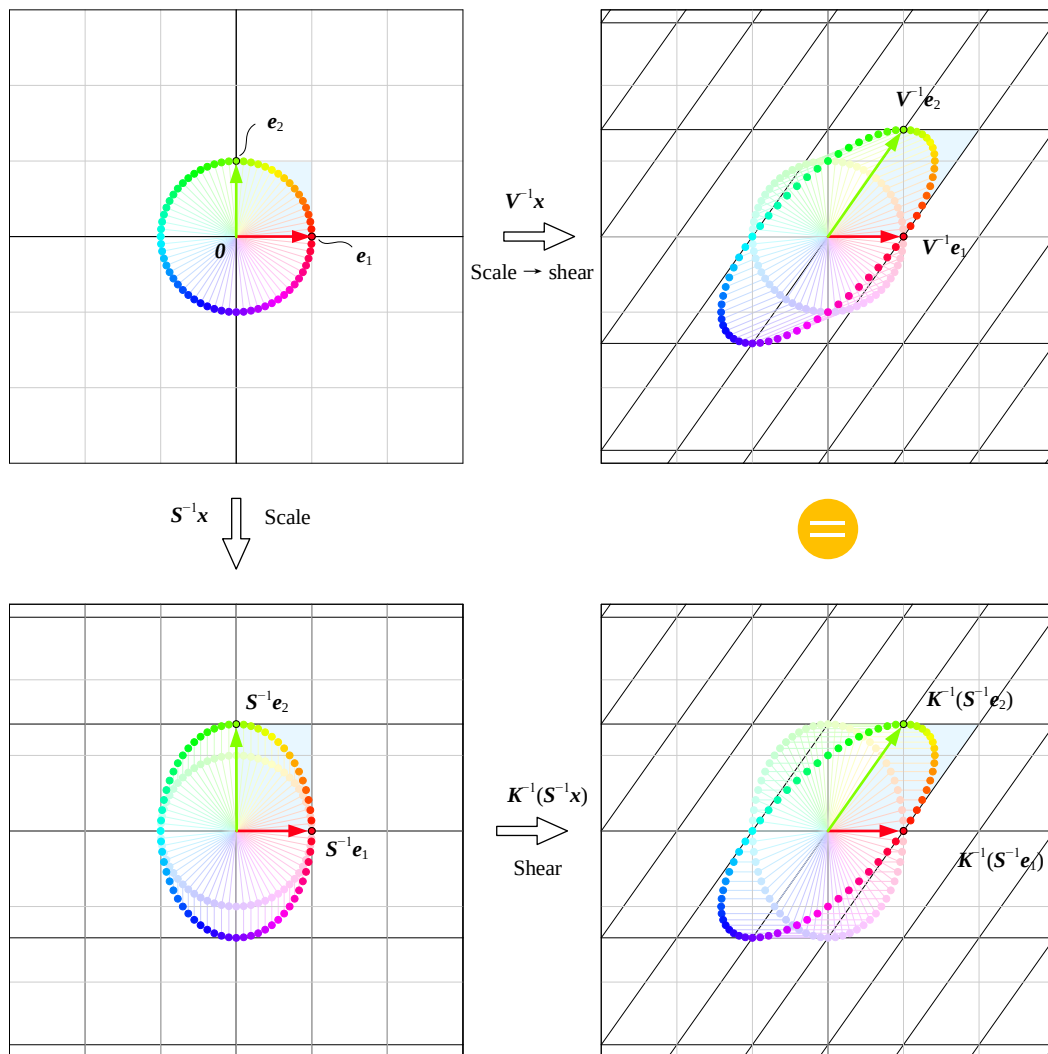


Figure 3. Linear Transformation of Matrix V^{-1} on e_1 and e_2 , Decomposed into First Scale (S^{-1}), Then Shear (K^{-1})

V 's inverse matrix, V^{-1} , has determinant

$$\det(V^{-1}) = \det\left(\begin{bmatrix} 1 & 1 \\ 0 & \sqrt{2} \end{bmatrix}\right) = \sqrt{2}$$

This means that in Figure 3, the area of the parallelogram grid is scaled by a factor of $\sqrt{2}$. Clearly, the shear matrix K^{-1} does not change the area of the shape; the scaling matrix S^{-1} is what causes the change in the shape's area.

V 's Action

As shown in Figure 4, V 's action is the opposite of V^{-1} : first shear (K), then scale (S).

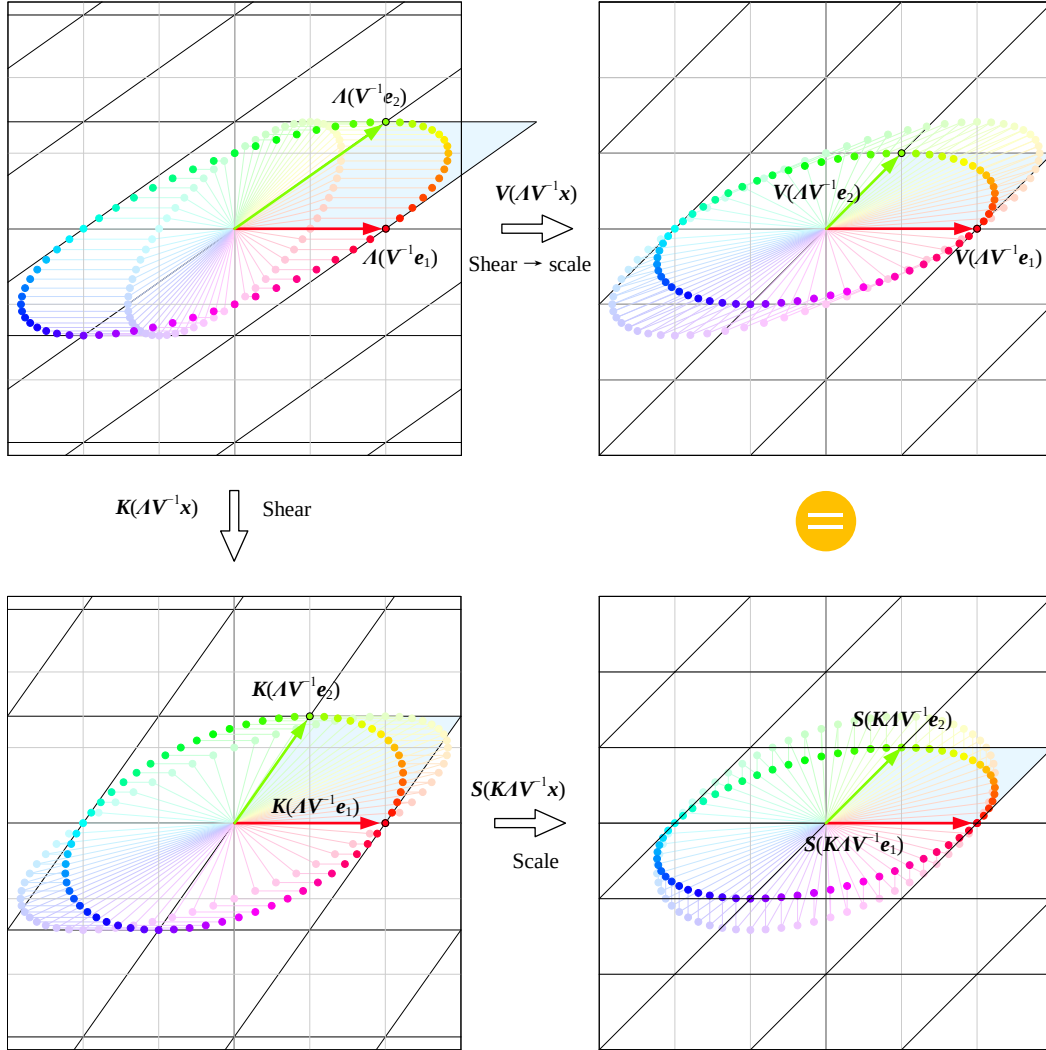


Figure 4. Linear Transformation of Matrix V , Decomposed into First Shear (K), Then Scale (S)

The determinant of matrix V is

$$\det(V) = \det\left(\begin{bmatrix} 1 & -\sqrt{2}/2 \\ 0 & \sqrt{2}/2 \end{bmatrix}\right) = \sqrt{2}/2$$

This means the area of the shape shrinks to $\sqrt{2}/2$ of its original size.

The determinants of V and its inverse multiply to 1. As mentioned earlier in this book, the product of the determinants of a matrix and its inverse always equals 1.

This means that the effects of V^{-1} and V on area cancel each other out, that is

$$\det(\mathbf{V}^{-1})\det(\mathbf{V})=1$$

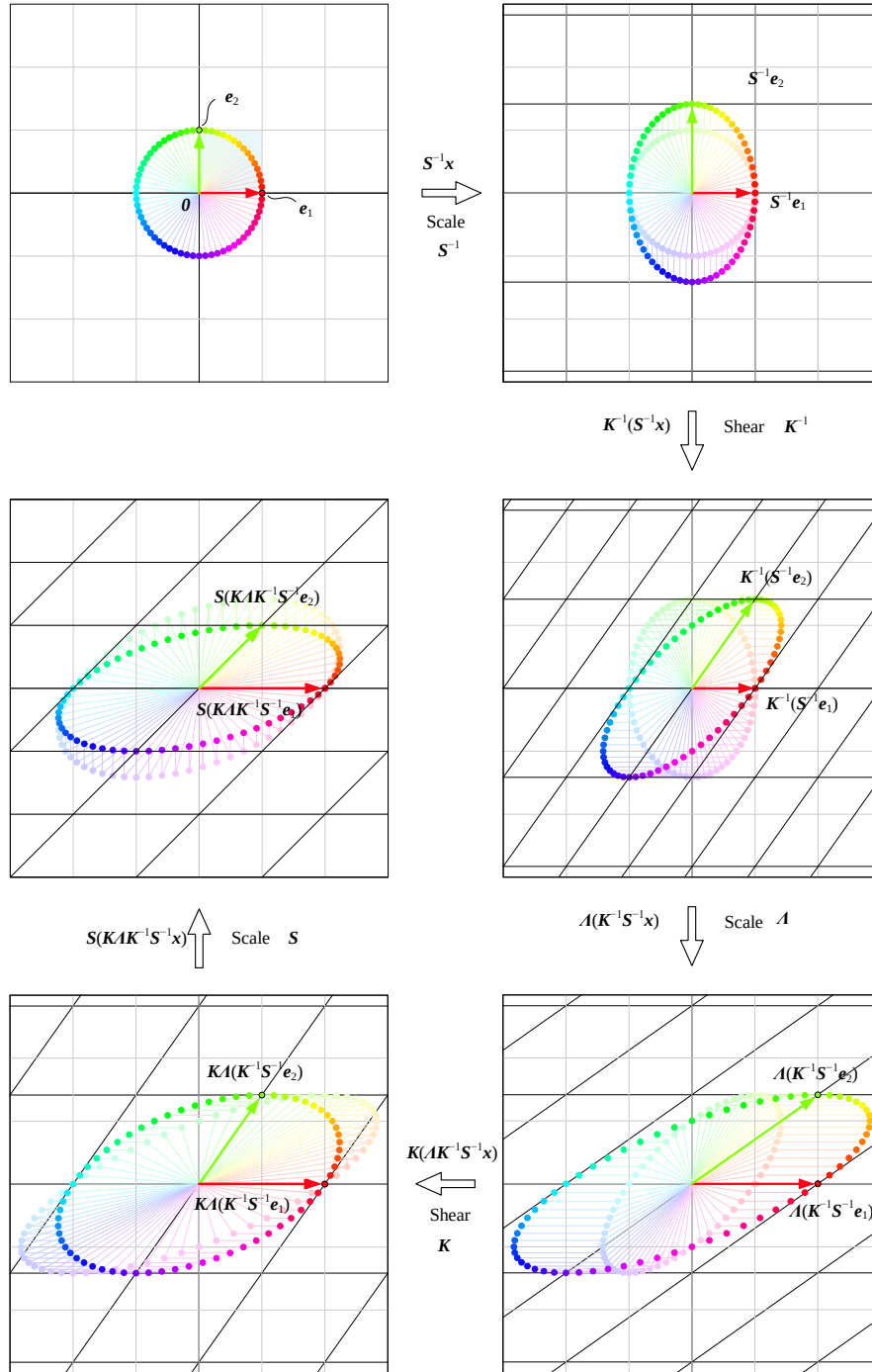
$\mathbf{A}$'s linear transformation causes the area change, coming from $\det(\mathbf{A})$, that is

$$\det(\mathbf{A}) = \det(\mathbf{A})$$

Scale $\rightarrow$ Shear $\rightarrow$ Scale $\rightarrow$ Shear $\rightarrow$ Scale

In other words, the eigenvalue decomposition of matrix $\mathbf{A}$ can be broken down into this sequence of actions: "scale ($\mathbf{S}^{-1}$) $\rightarrow$ shear ($\mathbf{K}^{-1}$) $\rightarrow$ scale ($\mathbf{A}$) $\rightarrow$ shear ($\mathbf{K}$) $\rightarrow$ scale ($\mathbf{S}$)", as shown specifically in Figure 5.

$$\mathbf{A} = \mathbf{V}\mathbf{A}\mathbf{V}^{-1} = \underbrace{\mathbf{S}\mathbf{K}}_{\mathbf{V}} \underbrace{\mathbf{A}\mathbf{K}^{-1}\mathbf{S}^{-1}}_{\mathbf{V}^{-1}}$$

Figure 5. "Scale $\rightarrow$ Shear $\rightarrow$ Scale $\rightarrow$ Shear $\rightarrow$ Scale" Linear Transformation on e_1 and e_2

The Non-symmetric Matrix Acting on v_1 and v_2

Next, let's look again at the geometric operation of the non-symmetric matrix A on the eigenvectors v_1 and v_2 .

As shown in Figure 6, the action of the non-symmetric matrix A on the eigenvector v_1 corresponds to the matrix multiplication Av_1

$$Av_1 = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix} = 2 \times \begin{bmatrix} 1 \\ 0 \end{bmatrix}_{v_1}$$

The action of the non-symmetric matrix A on the eigenvector v_2 corresponds to the matrix multiplication Av_2

$$Av_2 = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} -\sqrt{2}/2 \\ \sqrt{2}/2 \end{bmatrix} = \begin{bmatrix} -\sqrt{2}/2 \\ \sqrt{2}/2 \end{bmatrix} = \underbrace{1}_{\lambda_2} \times \begin{bmatrix} -\sqrt{2}/2 \\ \sqrt{2}/2 \end{bmatrix}_{v_2}$$

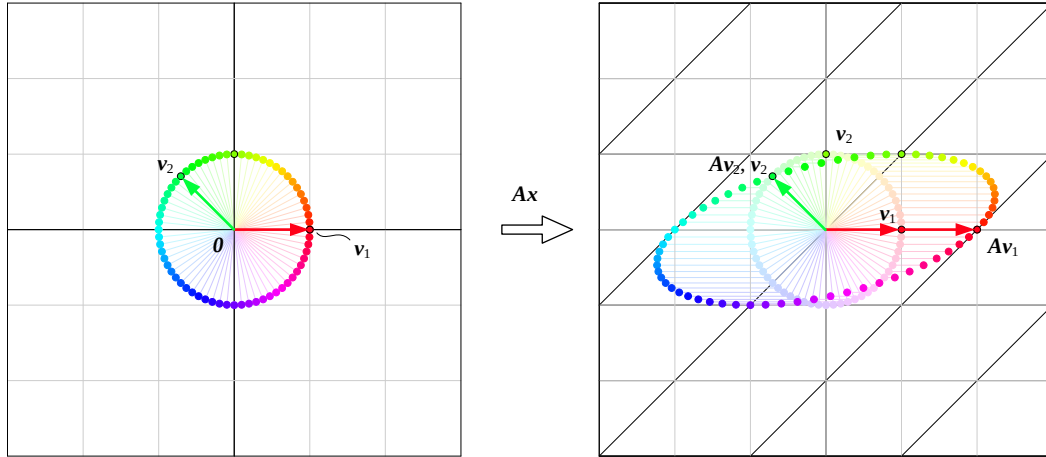


Figure 6. Linear Transformation of Matrix A on v_1 and v_2

Figure 7 shows the step-by-step linear transformation of the matrix product VAV^{-1} on v_1 and v_2 .



Please analyze the step-by-step operations in Figure 7 on your own.

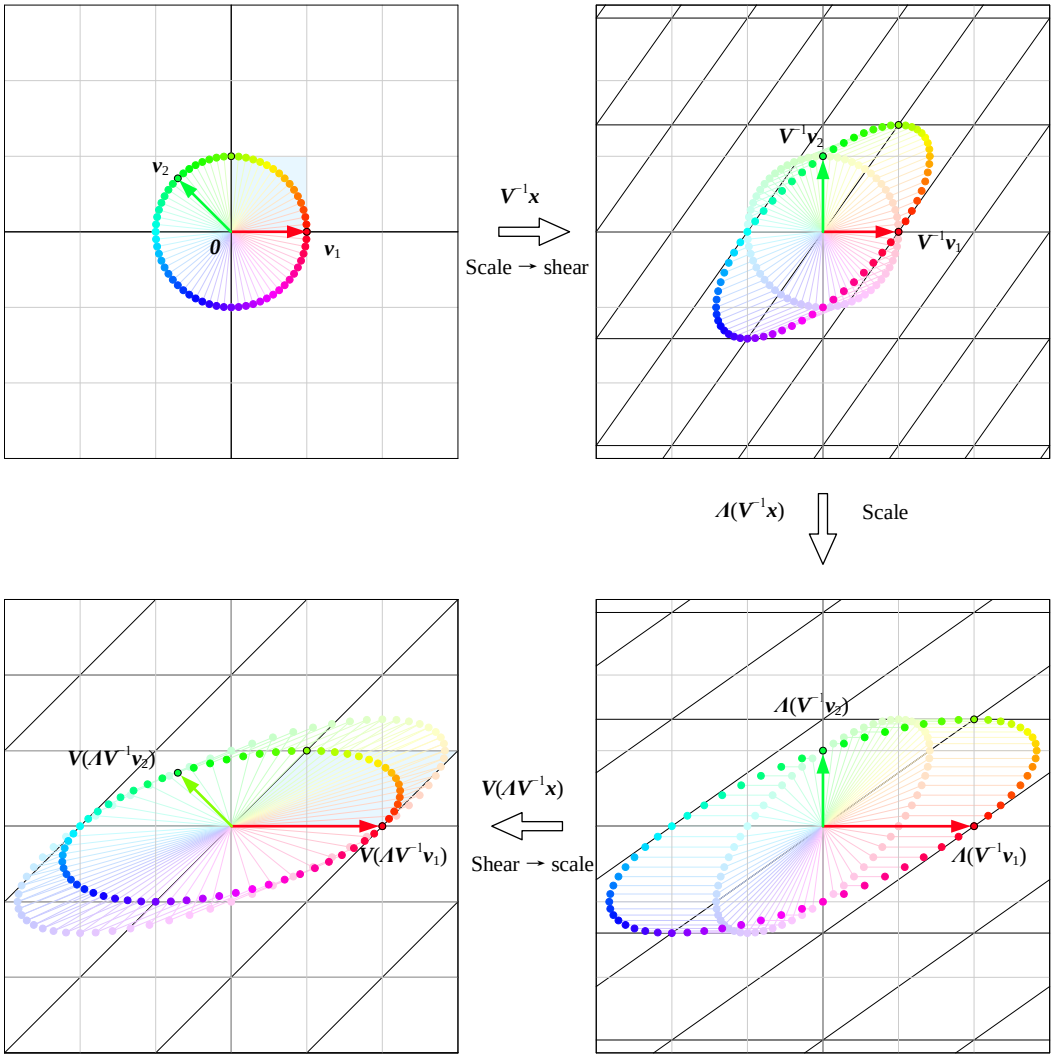
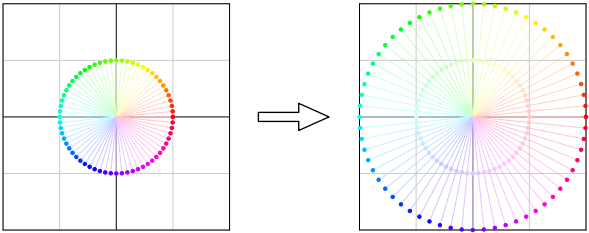


Figure 7. Linear Transformation of the Matrix Product AV^{-1} on v_1 and v_2

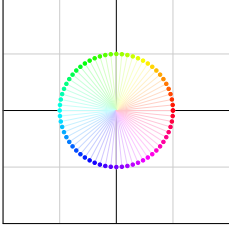
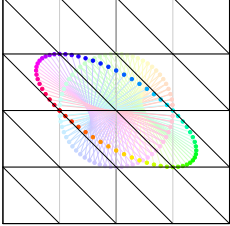
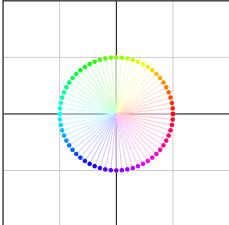
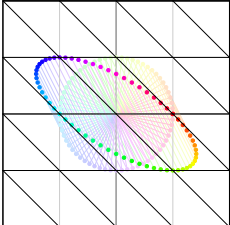
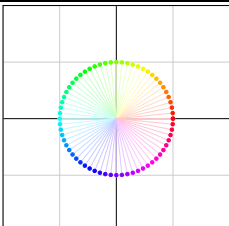
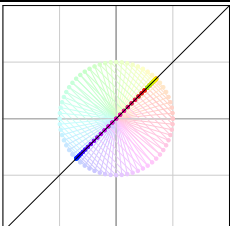
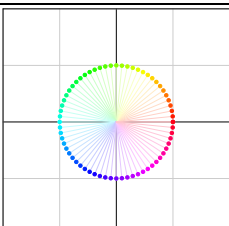
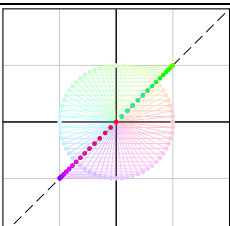
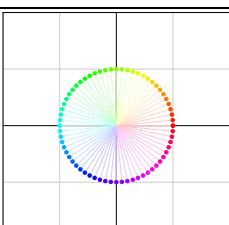
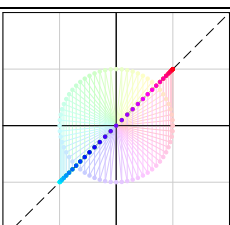
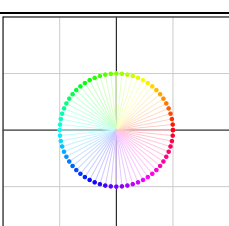
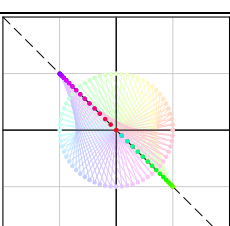
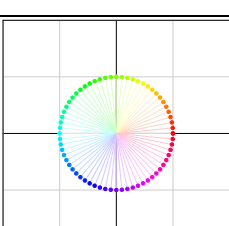
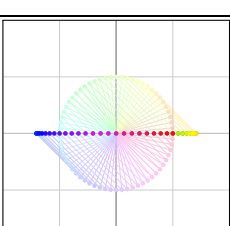
More Eigenvalue Decompositions

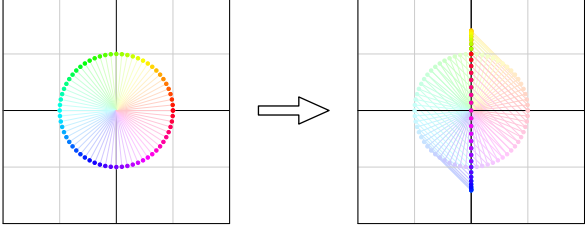
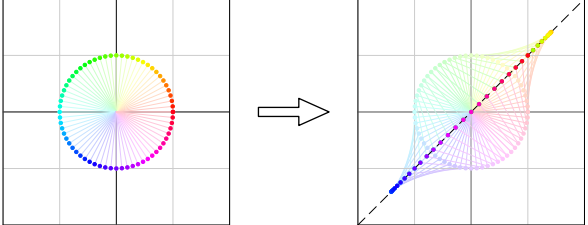
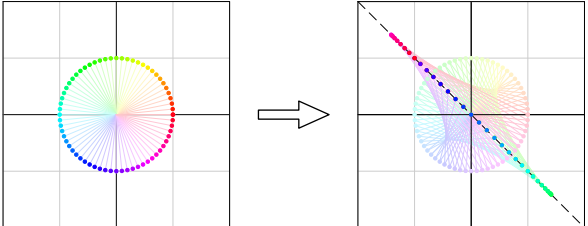
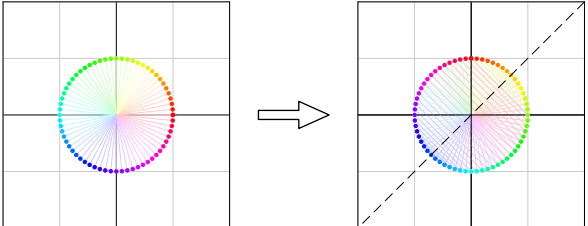
What was described above is just one example of the geometric perspective on the eigenvalue decomposition of a non-symmetric square matrix. Eigenvalue decomposition can produce many other interesting results – please analyze each one in Table 1.

Table 1. More Eigenvalue Decompositions

Matrix A	Eigenvalues, Eigenvectors	Geometric Transformation
$\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$ Can be diagonalized (already a diagonal matrix)	$\lambda_{1,2} = 2$ $v_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, v_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ Note: any nonzero vector can serve as an eigenvector	

$\begin{bmatrix} 1/2 & 0 \\ 0 & 2 \end{bmatrix}$ Can be diagonalized (already a diagonal matrix)	$\lambda_1 = \frac{1}{2}, \lambda_2 = 2$ $\mathbf{v}_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$	
$\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$ If $\sin \theta = 0$, the eigenvalues are real. The matrix can be diagonalized; the eigenvalues are +1 and -1, with two linearly independent eigenvectors	$\lambda_{1,2} = \cos \theta \pm i \sin \theta$ $\mathbf{v}_1 = \begin{bmatrix} 1 \\ -i \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 1 \\ +i \end{bmatrix}$	
$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ Cannot be diagonalized	$\lambda_{1,2} = 1$ $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$	
$\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ Cannot be diagonalized	$\lambda_{1,2} = 1$ $\mathbf{v}_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$	
$\begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$ Cannot be diagonalized	$\lambda_{1,2} = 1$ $\mathbf{v}_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$	
$\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$ Cannot be diagonalized	$\lambda_{1,2} = 1$ $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$	

$\begin{bmatrix} -1 & -1 \\ 0 & -1 \end{bmatrix}$ Cannot be diagonalized	$\lambda_{1,2} = -1$ $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$	 
$\begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix}$ Can be diagonalized	$\lambda_1 = 1, \lambda_2 = -1$ $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} -\sqrt{5}/5 \\ 2\sqrt{5}/5 \end{bmatrix}$	 
$\begin{bmatrix} 1 & 1 \\ 2 & 2 \\ 1 & 1 \\ 2 & 2 \end{bmatrix}$ Can be diagonalized	$\lambda_1 = 0, \lambda_2 = 1$ $\mathbf{v}_1 = \begin{bmatrix} \sqrt{2}/2 \\ \sqrt{2}/2 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} -\sqrt{2}/2 \\ \sqrt{2}/2 \end{bmatrix}$	 
$\begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}$ Can be diagonalized	$\lambda_1 = 0, \lambda_2 = 1$ $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} \sqrt{2}/2 \\ \sqrt{2}/2 \end{bmatrix}$	 
$\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$ Can be diagonalized	$\lambda_1 = 0, \lambda_2 = 1$ $\mathbf{v}_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} \sqrt{2}/2 \\ \sqrt{2}/2 \end{bmatrix}$	 
$\begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix}$ Can be diagonalized	$\lambda_1 = 0, \lambda_2 = -1$ $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} -\sqrt{2}/2 \\ \sqrt{2}/2 \end{bmatrix}$	 
$\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$ Can be diagonalized	$\lambda_1 = 1, \lambda_2 = 0$ $\mathbf{v}_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} -\sqrt{2}/2 \\ \sqrt{2}/2 \end{bmatrix}$	 

$\begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$ Can be diagonalized	$\lambda_1 = 1, \lambda_2 = 0$ $\mathbf{v}_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} \sqrt{2}/2 \\ -\sqrt{2}/2 \end{bmatrix}$	
$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ Can be diagonalized	$\lambda_1 = 2, \lambda_2 = 0$ $\mathbf{v}_1 = \begin{bmatrix} \sqrt{2}/2 \\ \sqrt{2}/2 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} -\sqrt{2}/2 \\ \sqrt{2}/2 \end{bmatrix}$	
$\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$ Can be diagonalized	$\lambda_1 = 0, \lambda_2 = -2$ $\mathbf{v}_1 = \begin{bmatrix} \sqrt{2}/2 \\ \sqrt{2}/2 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} -\sqrt{2}/2 \\ \sqrt{2}/2 \end{bmatrix}$	
$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ Can be diagonalized	$\lambda_1 = 1, \lambda_2 = -1$ $\mathbf{v}_1 = \begin{bmatrix} \sqrt{2}/2 \\ \sqrt{2}/2 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} -\sqrt{2}/2 \\ \sqrt{2}/2 \end{bmatrix}$	



Please use tools such as DeepSeek/ChatGPT to complete the following exercises for this section.

Q1. Please compute the eigenvalue decomposition of every square matrix in Table 1 by hand and with Python code, and analyze the results one by one. When analyzing each case in Table 1, please pay special attention to the following aspects

- Whether the square matrix is invertible (whether its determinant is 0);
- What the characteristic equation is;
- The properties of the eigenvalues – are they 0? Are they negative? Do complex numbers appear?
- Whether the eigenvectors are linearly dependent; if they are linearly independent, then $\mathbf{V}$ is invertible (the square matrix is diagonalizable).